ITA03-MOBILE COMPUTING

ASSIGNMENT-1

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual

Duration: 1 Week

Total Marks: 50

NAME:K.BALAJI

REG NO:192324180

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Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

● Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon). ● Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: $\leq 2\%$ call blocking.

● Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).

● Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.

● Spectrum cannot be increased; only the network layout and radio parameters may change.

● Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks Guiding Enquiry

1. Introduction

The university township covers 12 km² and contains academic buildings, hostels, residential areas and a main road. The cellular operator has a fixed allocation of 336 duplex voice channels.

The objective is to share these channels among multiple cells so that all subscribers receive reliable cellular service without excessive co-channel interference.

Three different frequency-planning strategies are proposed:

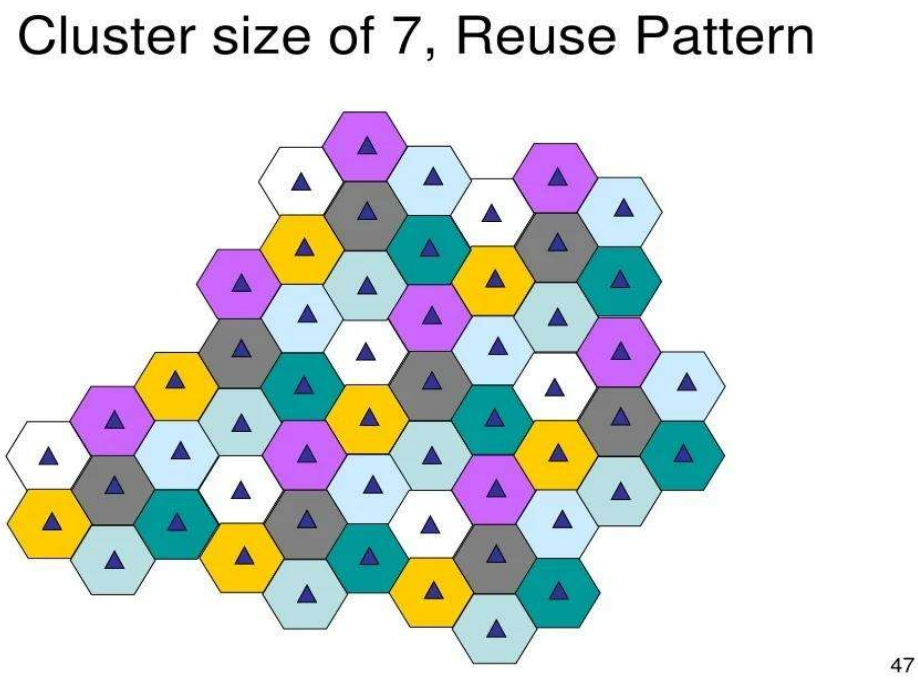
1. Fixed frequency reuse.
2. Sectorised frequency reuse.
3. Hierarchical macro–micro network with dynamic channel allocation.

2. Strategy 1: Fixed Frequency Reuse

Basic idea

In fixed frequency reuse, the entire coverage area is divided into hexagonal cells. The available channels are divided into groups and assigned to cells according to a repeating cluster pattern.

The same frequency group can be reused in cells that are sufficiently far apart to maintain acceptable signal quality.



The frequency reuse distance is determined by the cluster size:

$$D = R\sqrt{3N}$$

Where:

- \(D\) = distance between co-channel cells.
- \(R\) = cell radius.
- \(N\) = number of cells in a cluster.

Assumptions

- The township is represented by a regular hexagonal grid.
- Each cell has one base station.
- The total available spectrum is fixed at 336 duplex channels.
- The same channel group is reused in cells belonging to different clusters.
- Subscriber demand and propagation conditions are estimated before assigning channels.

Channel-sharing method

The 336 channels are divided into \(N\) groups.

$$C_{\text{cell}} = \frac{336}{N}$$

For example, with \(N=7\):

$$C_{\text{cell}} = \frac{336}{7} = 48$$

Each cell receives 48 duplex channels, assuming an exact division. The same set of 48 channels is reused in other clusters.

Iterative procedure

The following algorithm searches valid cluster sizes and selects a suitable frequency reuse pattern.

START

Input: 336 channels, cell radius R,

minimum SIR threshold

Generate a hexagonal cell grid

For each valid cluster size N:

Calculate reuse distance D = R * sqrt(3*N)

Divide 336 channels into N groups

Assign channel groups to cells

Calculate SIR for co-channel cells

If SIR >= minimum threshold:

Store the cluster size and allocation

Select the feasible cluster size

with the best capacity and interference trade-off

END

Valid cluster sizes are calculated using:

$$N = i^2 + ij + j^2$$

where \(i\) and \(j\) are non-negative integers.

Design principle

Spatial reuse.

Justification

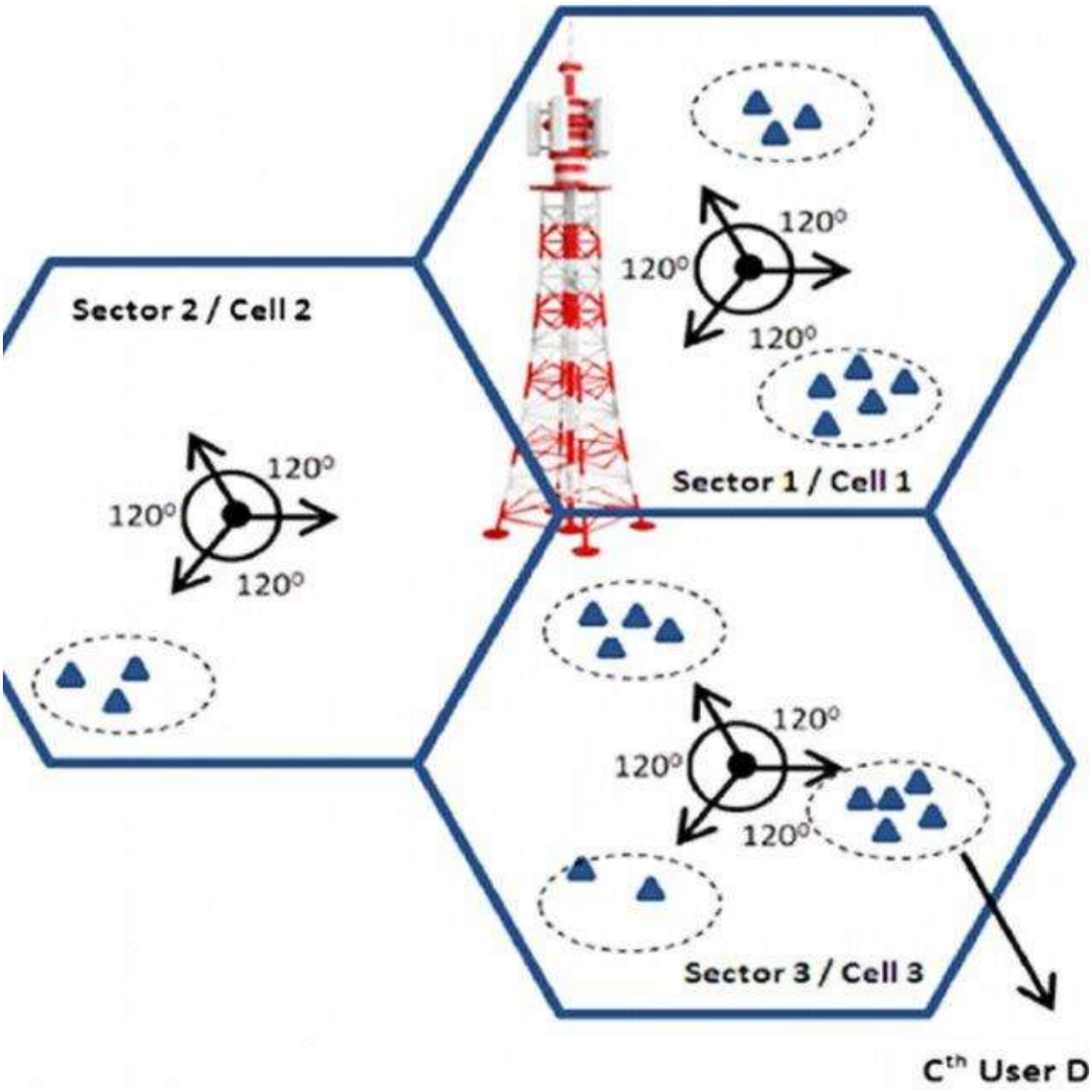
This strategy is simple to implement, requires predictable channel allocation and provides a useful baseline for comparing other strategies. However, a larger cluster reduces interference but also reduces the number of channels available per cell.

3. Strategy 2: Sectorised Frequency Reuse

Basic idea

In sectorised frequency reuse, each hexagonal cell is divided into three 120° sectors using directional antennas.

Each sector serves mobile users within its coverage direction. Directional antennas reduce interference from other co-channel cells because they transmit and receive signals mainly within their own sector.



Assumptions

- Each cell is divided into three 120° sectors.
- Each sector uses a directional antenna.
- The total spectrum remains fixed at 336 duplex channels.
- Antenna direction and sector boundaries are planned to minimise interference.
- Users are distributed among the three sectors.

Channel-sharing method

The 336 channels are divided into \(N\) frequency groups, and each cell is divided into three sectors.

For \(N=7\):

$$C_{\text{cell}} = \frac{336}{7} = 48$$

Each sector receives:

$$C_{\text{sector}} = \frac{48}{3} = 16$$

The three sectors together use 48 channels per cell.

Sectorisation reduces the number of dominant co-channel interferers from six in an ideal unsectorised first-tier model to approximately two per 120° sector.

START

Input: 336 channels, cluster size N

Generate a hexagonal cell grid

Divide each cell into three sectors

Divide the available channels into N groups

For each cell:

Assign a frequency group

For each sector:

Assign one-third of the cell channels

Set antenna direction

Calculate co-channel SIR

Check whether the SIR meets

the minimum acceptable threshold

If acceptable:

Store the sectorised layout

END

Design principle

Spatial filtering.

Justification

Sectorisation improves interference management and allows more efficient frequency reuse. It is particularly useful in dense areas such as academic buildings, hostels and the university's main road.

However, it requires additional directional antennas, careful antenna orientation and more complex handoff management.

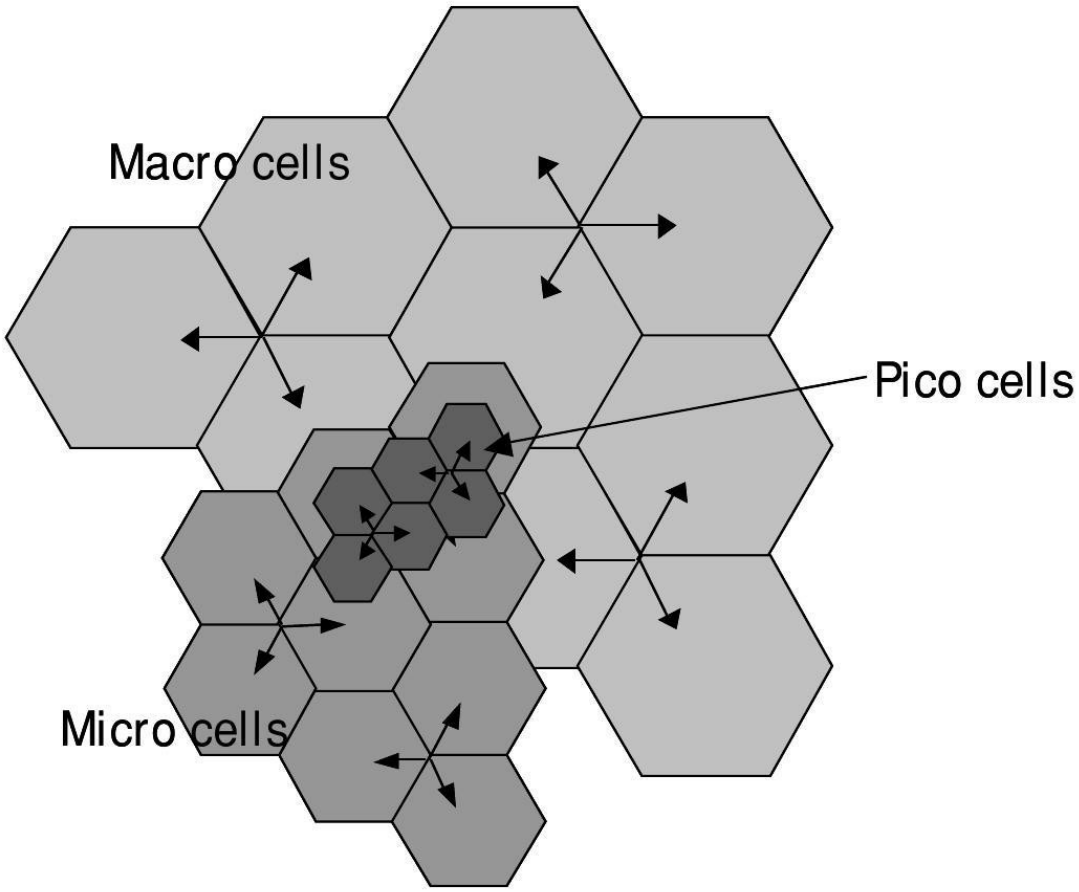
4. Strategy 3: Hierarchical Macro–Micro Network

Basic idea

A hierarchical network consists of large macrocells and smaller microcells.

Macrocells provide wide-area coverage, while microcells are deployed in high-demand locations such as stadiums, academic blocks and crowded hostels.

When a cell becomes overloaded, it is divided into smaller cells through a process called cell splitting.



Assumptions

- Macrocells provide the initial coverage.
- Microcells are introduced only in overloaded areas.
- The total available spectrum remains fixed at 336 duplex channels.
- Smaller cells use lower transmit power.

- Channel allocation is adjusted according to traffic demand and interference.

Channel-sharing method

The macrocell layer provides basic coverage, while microcells provide additional capacity in areas with high traffic.

Channels can be borrowed dynamically from neighbouring cells when demand increases, provided that the reuse and interference constraints are satisfied.

The number of channels available to each cell is therefore not necessarily fixed.

Recursive procedure

START

Input: Initial cell radius R,

traffic demand and available channels

Create the initial macrocell network

For each cell:

Measure traffic demand

If demand exceeds available capacity:

Split the cell into four smaller cells

Reduce the cell radius to R/2

Recalculate channel allocation

Recalculate interference

Repeat the procedure for

any overloaded smaller cells

Else:

Retain the existing cell

END

Design principle

Hierarchical spatial reuse and dynamic trunking.

Justification

This strategy provides a flexible way to increase capacity without purchasing additional spectrum. It is especially useful for the stadium event, where traffic demand is high for only a few hours.

However, it requires additional base stations, careful channel coordination and increased handoff management.

5. Comparison of the three strategies

Feature	Fixed reuse	Sectorised reuse	Macro–micro
Main principle	Spatial reuse	Spatial filtering	Hierarchical reuse
Channel allocation	Fixed	Fixed by sector	Dynamic or demand-based
Interference control	Reuse distance	Directional antennas	Smaller cells and reuse
Capacity improvement	Limited by cluster size	Improved interference management	High in overloaded areas
Main cost	Initial planning	Directional antennas	Additional sites and planning

6. Preprocessing requirements and cost

Before implementing any of these strategies, the following preprocessing is required.

Requirement	Purpose	Cost bearer
Traffic map	Identify areas with high subscriber demand	Cellular operator
Propagation survey	Estimate path loss, coverage and interference	Cellular operator
Hexagonal grid layout	Determine cell locations and reuse distances	Cellular operator
Channel allocation plan	Assign frequencies and prevent interference	Cellular operator

The operator bears these planning costs because it is responsible for providing cellular coverage to the university township.

The exact cost depends on the number of survey locations, base stations and planning resources required.

7. Access channels and their role

Access channels allow mobile devices to communicate with the cellular network.

- FDMA: Divides the available spectrum into separate frequency channels.
- TDMA: Divides a frequency channel into time slots shared by different users.
- CDMA: Allows users to share the same frequency band using different spreading codes.

- OFDMA: Divides the spectrum into orthogonal subcarriers that can be allocated to users.

Control channels are used for signalling, registration, paging and call setup. Traffic channels carry voice or data communication.

The given 336 channels are already specified after control-channel overhead, so the calculations in this assignment use 336 as the available traffic-channel allocation.

TASK 2 – Identify the Governing Design Parameter 2 Marks Guiding Enquiry

1. Introduction

The governing design parameter is the most important quantity that determines the capacity and interference performance of a cellular network.

In this assignment, three frequency-planning strategies are considered: fixed frequency reuse, sectorised frequency reuse and hierarchical macro–micro networking.

Each strategy has a different governing parameter because each controls frequency sharing and interference in a different way.

2. Governing parameter for each strategy

Strategy	**Governing parameter**	**Reason**
Fixed frequency reuse	Cluster size, N	Determines the number of channels per cell and the distance between co-channel cells.
Sectorised frequency reuse	Number of sectors, S	Determines the number of dominant co-channel interferers and the interference reduction achieved by directional antennas.
Hierarchical macro–micro network	Cell radius, R	Determines the coverage area, number of cells, capacity per unit area and handoff frequency.

3. Strategy 1 – Fixed Frequency Reuse

The governing parameter is the cluster size \(N\).

The available channels are divided among the cells in a cluster.

$$C_{\text{cell}} = \frac{S}{N}$$

Where:

- \(S\) = total number of available duplex channels.

- \(N\) = number of cells in a cluster.
- \(C_{\text{cell}}\) = channels available per cell.

For the given 336 channels:

$$C_{\text{cell}} = \frac{336}{N}$$

As the cluster size increases, the number of channels per cell decreases. However, the reuse distance increases, reducing co-channel interference.

The reuse distance is:

$$D = R\sqrt{3N}$$

Therefore, cluster size controls the trade-off between capacity and interference.

When subscriber demand increases, a smaller cluster size may be preferred to increase channel availability, provided the minimum SIR requirement is satisfied.

4. Strategy 2 – Sectorised Frequency Reuse

The governing parameter is the number of sectors per cell, denoted by \(S_{\text{sec}}\).

For three-sector cells:

$$S_{\text{sec}} = 3$$

Each sector covers 120° of the cell.

The number of dominant first-tier co-channel interferers is reduced from approximately six in an ideal unsectorised model to approximately two per sector.

The SIR is approximately:

$$\frac{SIR_{\text{sector}}}{SIR_{\text{omni}}} \approx \frac{6}{2} = 3$$

In decibels:

$$10\log_{10}(3) \approx 4.77 \text{ dB}$$

\[10\log_{10}(3)\approx4.77\text{ dB} \]

Thus, sectorisation can improve the SIR by approximately 4.77 dB under the simplified equal-distance interference model.

Increasing the number of sectors can improve spatial filtering, but it also increases antenna and handoff complexity.

When subscriber demand increases, sectorisation can help improve capacity by reducing interference and enabling more aggressive frequency reuse.

5. Strategy 3 – Hierarchical Macro–Micro Network

The governing parameter is the cell radius \(R\).

The area of a hexagonal cell is:

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2}R^2$$

The number of cells required to cover the township is:

$$N_{\text{cells}} = \frac{12}{\frac{3\sqrt{3}}{2}R^2}$$

Therefore:

$$N_{\text{cells}} \propto \frac{1}{R^2}$$

As the cell radius decreases, more cells are required to cover the same area.

When the radius is reduced by half:

$$R' = \frac{R}{2}$$

The coverage area of each cell becomes one-fourth of its original area, and approximately four times as many cells are required.

This increases spatial reuse and allows the network to support more subscribers. However, it also increases the number of base stations and handoffs.

When demand increases, smaller cells can be introduced in overloaded areas to increase capacity without increasing the available spectrum.

6. Interpretation for subscriber growth

The township subscriber count grows from 1,000 to 100,000 or more.

Subscribers	**Offered traffic**
1,000	25 Erlangs
10,000	250 Erlangs
20,000	500 Erlangs
100,000	2,500 Erlangs

The offered traffic increases linearly with the number of subscribers.

As demand increases:

- Fixed frequency reuse requires careful cluster-size selection to balance channel capacity and interference.
- Sectorised reuse can improve interference management and channel efficiency.
- Hierarchical networking requires smaller cells in high-demand areas to increase capacity.

Task 3 – Derive the Capacity and Interference Functions Mathematically

1. Introduction

The objective of this task is to derive the mathematical relationships between hexagonal cell geometry, frequency reuse, network capacity, co-channel interference and planning cost.

The given planning data are:

Parameter	**Value**
Township area	12 km²
Available duplex traffic channels	336
Busy-hour traffic per subscriber	0.025 Erlang
Target call blocking probability	2%
Minimum acceptable SIR	12–18 dB
Path-loss exponent	2.5–4

The calculations below use a regular hexagonal grid, uniform channel allocation and the ideal first-tier interference model. Where necessary, assumptions are stated explicitly.

2. Hexagonal geometry and frequency reuse

2.1 Hexagonal cell area

Let \(R\) be the distance from the centre of a hexagonal cell to one of its vertices.

The hexagon consists of six equilateral triangles.

The area of one triangle is:

$$A_{\triangle} = \frac{1}{2} R^2 \sin 60^\circ$$

$$A_{\triangle} = \frac{\sqrt{3}}{4} R^2$$

Therefore, the area of one hexagonal cell is:

$$A_{\text{cell}} = 6A_{\triangle}$$

$$\boxed{A_{\text{cell}} = \frac{3\sqrt{3}}{2}R^2}$$

For a cell radius of 0.5 km:

$$A_{\text{cell}} = \frac{3\sqrt{3}}{2}(0.5)^2$$

$$A_{\text{cell}} \approx 0.6495 \text{ km}^2$$

2.2 Number of cells required

The township area is 12 km².

The approximate number of cells required is:

$$M = \frac{A_{\text{township}}}{A_{\text{cell}}}$$

$$M = \frac{12}{\frac{3\sqrt{3}}{2}R^2}$$

$$\boxed{M \approx \frac{4.619}{R^2}}$$

For \(R=0.5\) km:

$$M = \frac{12}{0.6495} \approx 18.48$$

Therefore, approximately 19 cells are required.

This is an area-based estimate; the actual number of cells in a finite hexagonal layout may differ because of boundary effects.

2.3 Cluster-size relationship

In a hexagonal grid, a co-channel cell can be located by moving \(i\) cells along one grid direction and then \(j\) cells along a direction 60° away.

The distance between the two cell centres is:

$$D^2 = R^2(3i^2 + 3ij + 3j^2)$$

Dividing by \(3R^2\):

$$\frac{D^2}{3R^2} = i^2 + ij + j^2$$

The cluster size is therefore:

$$\boxed{N = i^2 + ij + j^2}$$

The reuse distance is:

$$\boxed{D = R\sqrt{3N}}$$

The reuse ratio is:

$$\boxed{Q = \frac{D}{R} = \sqrt{3N}}$$

For \(N=7\):

$$Q = \sqrt{21} \approx 4.583$$

Hence, the co-channel reuse distance is approximately 4.583 times the cell radius.

3. Channel capacity and traffic calculations

3.1 Channels per cell

Let:

- \(S=336\) be the total number of available duplex traffic channels.
- \(N\) be the frequency reuse cluster size.
- \(C\) be the number of channels per cell.

For fixed frequency reuse:

\[C =\$frac{S}{N} \]

For example, with \(N=7\):

$$C = \frac{336}{7} = 48 \text{ channels/cell}$$

For a cluster size of 3, 4, 7 and 12, the corresponding channel allocations are 112, 84, 48 and 28 channels per cell, respectively.

3.2 Offered traffic

Offered traffic is calculated using:

$$A = \lambda H$$

where \(\lambda\) is the average call arrival rate and \(H\) is the average call holding time.

Given that each subscriber generates 0.025 Erlang during the busy hour:

$$A_{\text{total}} = 0.025U$$

where \(U\) is the number of subscribers.

For 1,000 subscribers:

$$A_{\text{total}} = 0.025(1000) = 25 \text{ Erlangs}$$

For 100,000 subscribers:

$$A_{\text{total}} = 0.025(100000) = 2500 \text{ Erlangs}$$

Assuming uniform traffic distribution, the offered traffic per cell is:

$$A_{\text{cell}} = \frac{0.025U}{M}$$

3.3 Erlang-B blocking probability

The Erlang-B formula determines the probability that an arriving call is blocked because all channels are occupied.

$$B(C, A) = \frac{\frac{A^C}{C!}}{\sum_{k=0}^{C} \frac{A^k}{k!}}$$

The target blocking probability is:

$$B(C, A) \leq 0.02$$

For a cell with 48 channels, the maximum offered traffic must be found by solving:

$$B(48, A_{\max}) = 0.02$$

This gives approximately 38 Erlangs of offered traffic per cell.

Thus, a 48-channel cell can support approximately:

$$U_{\text{cell}} = \frac{38}{0.025}$$

$$\boxed{U_{\text{cell}} \approx 1520 \text{ subscribers}}$$

This is a theoretical estimate under the Erlang-B assumptions of independent call arrivals, exponential call durations and no waiting queue.

3.4 Total system capacity

The approximate total offered traffic capacity is:

$$A_{\text{system}} = MA_{\max}$$

The total subscriber capacity is:

$$U_{\max} = \frac{MA_{\max}}{0.025}$$

For a fixed reuse cluster:

$$\boxed{U_{\max} = \frac{MA_{\max}(336/N)}{0.025}}$$

Here, \(A_{\max}(C)\) is the maximum offered traffic that satisfies the 2% blocking target for \(C\) channels.

The actual capacity also depends on whether the SIR requirement is satisfied.

4. Co-channel interference and SIR

4.1 Basic SIR expression

The signal-to-interference ratio is:

$$SIR = \frac{P_r}{I}$$

where \(P_r\) is the received power from the serving base station and \(I\) is the total received interference power from co-channel base stations.

Assuming a path-loss model:

\[P_r\propto d^{-n} \]

where \(d\) is the distance from the transmitting base station and \(n\) is the path-loss exponent.

For \(i_0\) dominant co-channel interferers:

\[SIR= \frac{R^{-n}} {\displaystyle\sum_{i=1}^{i_0}D_i^{-n}} \]

where \(D_i\) is the distance from the mobile to interferer \(i\).

4.2 Fixed frequency reuse

For a mobile at the cell boundary, assume six first-tier interferers, each approximately \(D\) away.

\[SIR=\frac{R^{-n}}{6D^{-n}} \]

\[SIR=\frac{1}{6}\left(\frac{D}{R}\right)^n \]

Using \(D/R=\sqrt{3N}\):

\[\boxed{SIR=\frac{(3N)^{n/2}}{6}} \]

In decibels:

\[\boxed{ SIR_{\text{dB}}= 10\log_{10}\left(\frac{(3N)^{n/2}}{6}\right) } \]

For \(N=7\) and \(n=4\):

\[SIR=\frac{21^2}{6}=73.5 \]

\[SIR_{\text{dB}}=10\log_{10}(73.5) \]

\[\boxed{SIR_{\text{dB}}\approx18.66\text{ dB}} \]

This agrees with the assignment's reference value of approximately 18.7 dB.

4.3 Sectorised frequency reuse

For ideal three-sector cells, assume two dominant interferers per sector instead of six.

\[SIR_{\text{sector}}= \frac{(3N)^{n/2}}{2} \]

The theoretical improvement over the unsectorised case is:

$$\frac{SIR_{\text{sector}}}{SIR_{\text{omni}}} = \frac{6}{2} = 3$$

\[10\log_{10}(3)\approx4.77\text{ dB} \]

Therefore, ideal sectorisation improves the SIR by approximately 4.77 dB.

This is an idealised result. Actual SIR depends on antenna beamwidth, sidelobes, sector orientation and mobile position.

4.4 Hierarchical macro–micro network

For a macrocell, let the radius be \(R\). After cell splitting:

\[R'=\frac{R}{2} \]

If the reuse cluster remains unchanged, the co-channel reuse distance also halves:

$$D' = \frac{D}{2}$$

The ideal SIR expression remains unchanged because the ratio \(D/R\) remains the same.

However, the smaller cell serves a smaller area, allowing additional base stations and more spatial reuse. Actual SIR depends on the locations of the new microcells, their transmit power and their frequency assignments.

5. Planning cost and capacity relationship

Let:

- \(c_s\) = cost of one site.
- \(c_p\) = fixed planning and survey cost.
- \(M\) = number of sites.
- \(U_{\max}\) = subscriber capacity.

The total planning cost is:

\[C_{\text{total}}=c_p+Mc_s \]

For fixed frequency reuse:

\[C_{\text{total}}= c_p+c_s\frac{12}{A_{\text{cell}}} \]

For a macro–micro network, the number of additional sites must be included:

$$C_{\text{total}} = c_p + c_s(M_{\text{macro}} + M_{\text{micro}})$$

The cost per supported subscriber is:

\[\boxed{ C_{\text{per subscriber}}= \frac{C_{\text{total}}}{U_{\max}} } \]

A design is economically efficient when it provides sufficient capacity and SIR at the lowest practical cost per subscriber.

The operator must balance the cost of additional sites against the improvement in capacity and interference performance.

Task 4 – Analyse Best, Average and Worst-Case Behaviour

1. Introduction

The performance of a cellular network changes depending on the position of the mobile user, the traffic load and the propagation environment.

The best case occurs when signal quality is high and interference is low. The worst case occurs when the received signal is weak, interference is high or the available channels are fully occupied.

The following analysis considers the three proposed frequency-planning strategies.

2. Best, average and worst SIR

For the calculations, use the following assumptions:

- Cluster size \(N=7\).
- Cell radius \(R=0.5\) km.
- Path-loss exponent \(n=4\).
- Six first-tier co-channel interferers for an unsectorised cell.
- Two dominant interferers per sector for an ideal three-sector cell.

The reuse distance is:

$$D = 0.5\sqrt{21} = 2.291 \text{ km}$$

2.1 Fixed frequency reuse

The SIR is calculated using:

\[SIR=\frac{R^{-n}}{\sum D_i^{-n}} \]

The following positions are used to represent the different cases.

Case	Mobile position	Expected behaviour
Best	Near the base station	Strong serving signal and relatively low interference
Average	Near the middle of the cell	Moderate serving signal and interference
Worst	Near the cell boundary facing co-channel interferers	Weak serving signal and potentially high interference

For a mobile at the cell centre, the serving distance approaches zero, so the idealised path-loss model predicts extremely high SIR. In practice, SIR is limited by antenna patterns, receiver noise and other physical effects.

For the cell-edge approximation, use the six-equidistant-interferer model:

$$SIR = \frac{(3N)^{n/2}}{6}$$

$$SIR = \frac{(21)^2}{6} = 73.5$$

$$SIR_{\text{dB}} = 18.66 \text{ dB}$$

This is the approximate worst-case SIR in the simplified model.

The actual worst case may be lower because the six interferers are not exactly equidistant from every cell-edge location.

2.2 Sectorised frequency reuse

For an ideal three-sector cell, two dominant interferers are assumed.

$$SIR_{\text{sector}} = \frac{(3N)^{n/2}}{2}$$

$$SIR_{\text{sector}} = \frac{21^2}{2} = 220.5$$

$$SIR_{\text{sector,dB}} = 10\log_{10}(220.5)$$

$$\boxed{SIR_{\text{sector,dB}} \approx 23.43 \text{ dB}}$$

The sectorised network provides approximately 4.77 dB improvement over the unsectorised network under the same ideal assumptions.

At the cell centre, the serving signal is strongest. Near a sector boundary, the mobile may receive interference from adjacent sectors and the SIR may decrease.

2.3 Hierarchical macro–micro network

For a macrocell of radius \(R=0.5\) km, the approximate number of cells required is:

\[M=\frac{12}{0.6495}\approx19 \]

After splitting a macrocell:

\[R'=\frac{R}{2}=0.25\text{ km} \]

The area of each smaller cell becomes:

$$A'_{\text{cell}} = \frac{3\sqrt{3}}{2}(0.25)^2$$

$$A'_{\text{cell}} \approx 0.1624 \text{ km}^2$$

Four smaller cells cover approximately the same area as one original cell.

If the reuse pattern is unchanged, the ideal SIR remains approximately 18.66 dB because both the cell radius and reuse distance are reduced by the same factor.

However, smaller cells may experience higher interference if their frequency assignments are not coordinated properly.

3. Best, average and worst blocking behaviour

Blocking probability depends on offered traffic and the number of available channels.

The Erlang-B formula is:

$$B(C, A) = \frac{A^C/C!}{\sum_{k=0}^{C} A^k/k!}$$

For 48 channels, the approximate maximum offered traffic at 2% blocking is 38 Erlangs.

Traffic conditions

Case	Traffic condition	Expected blocking
Best	Off-peak traffic	Low blocking
Average	Normal busy-hour traffic	Moderate blocking
Worst	Event surge or unusually high demand	High blocking

For a cell with 48 channels, consider these illustrative traffic loads:

- Off-peak: \(A=10\) Erlangs.
- Average busy hour: \(A=30\) Erlangs.
- Overloaded cell: \(A=50\) Erlangs.

The Erlang-B probabilities are approximately:

Offered traffic	Blocking probability

10 Erlangs	0.00002%
30 Erlangs	0.43%
50 Erlangs	8.0%

The 50-Erlang case exceeds the target of 2% blocking.

This shows that even when SIR is acceptable, the network may fail to meet the required grade of service if too many users compete for the same channels.

4. Effect of shadowing and path-loss exponent

Shadowing occurs when buildings, trees or other obstacles reduce the received signal strength.

The received power follows:

\[P_r\propto d^{-n} \]

For \(n=2.5\), the signal decreases more slowly with distance than for \(n=4\).

A higher path-loss exponent reduces interference from distant co-channel cells, but it also causes the serving signal to weaken more rapidly as the mobile moves away from its base station.

For the \(N=7\) cluster:

$$SIR = \frac{(21)^{n/2}}{6}$$

At \(n=2.5\):

$$SIR_{\text{dB}} \approx 7.85 \text{ dB}$$

At \(n=4\):

$$SIR_{\text{dB}} \approx 18.66 \text{ dB}$$

Thus, the ideal SIR is higher in the dense built-up environment because distant interferers are attenuated more strongly.

However, the dense environment may have additional shadowing and obstruction losses. Therefore, the real network must be evaluated using propagation surveys and simulation rather than relying only on the path-loss exponent.

TASK 5 – Scaling and Sensitivity Analysis (3 Marks)

1. Aim

To investigate how network capacity, interference, and the number of cells change when the number of subscribers and environmental conditions increase.

2. Scaling relationships

A. Capacity and frequency reuse

The number of channels allocated to each cell is:

$$C_{\text{cell}}=\frac{S}{N}$$

Where:

- S = Total available channels.
- N = Frequency reuse factor.

Therefore,

$$\boxed{C_{\text{cell}}\propto\frac{1}{N}}$$

As N increases, the number of channels available per cell decreases.

B. Signal-to-interference ratio (SIR)

For a hexagonal cellular network:

$$\frac{D}{R}=\sqrt{3N}$$

The approximate SIR is:

$$SIR=\frac{(3N)^{n/2}}{i_0}$$

Where:

- D = Co-channel reuse distance.
- R = Cell radius.
- n = Path-loss exponent.
- i_0 = Number of dominant interfering cells.

For six dominant interferers:

$$SIR=\frac{(3N)^{n/2}}{6}$$

Therefore,

$$\boxed{SIR\propto N^{n/2}}$$

Increasing N improves SIR but reduces the number of channels per cell.

C. Number of cells

The approximate cell coverage area is:

$$A=\frac{3\sqrt{3}}{2}R^2$$

For a fixed geographical area:

$$\boxed{\text{Number of cells}\propto\frac{1}{R^2}}$$

If the radius is reduced by half:

$$\frac{1}{(R/2)^2}=\frac{4}{R^2}$$

Therefore, approximately four times as many cells are required.

3. Sensitivity analysis

Effect of changing the reuse factor

Reuse factor N	Channels per cell (relative)	Interference
3	1/3	High
4	1/4	Moderate
7	1/7	Low
12	1/12	Very low

Effect of path-loss exponent

For N = 7 and six dominant interferers:

Path-loss exponent	Approximate SIR
2.5	12.9 dB
3.0	15.8 dB
3.5	18.5 dB
4.0	21.0 dB

A higher path-loss exponent reduces the received interference more rapidly because distant interfering signals become weaker.

4. Subscriber growth interpretation

Subscribers	Network condition	Suitable strategy
1,000	Low traffic	Conventional macro cells
10,000	Moderate traffic	Frequency reuse optimization
20,000	Increasing traffic	Sectoring and selective splitting
100,000	High traffic	Cell splitting and small cells
Above 100,000	Very high traffic	Heterogeneous network with microcells

5. Break-even analysis

A larger reuse factor improves interference but reduces channel availability.

For example, moving from N = 3 to N = 7 reduces relative channels per cell from 1/3 to 1/7.

Cell splitting becomes preferable when the traffic demand exceeds the capacity of existing cells and the additional infrastructure cost is justified by the increased capacity.

Conclusion:

Frequency reuse is suitable for controlling interference, while cell splitting is more suitable for increasing capacity. For a township growing towards 100,000 subscribers, a combination of frequency reuse, sectoring and selective cell splitting provides a better solution than relying on only one strategy.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

1. Aim

To formulate recurrence relations for the number of cells, network capacity, transmit power and handoff load when overloaded cells are repeatedly split.

2. Original problem

Assume the initial network contains:

- 10 cells.
- 1,000 subscribers.
- Cell radius = 1 km.
- Initial capacity = 1,000 subscribers.

When a cell becomes overloaded, it is split into four smaller cells.

The new radius becomes:

$$R_k = \frac{R_{k-1}}{2}$$

3. Number of new cells

One original cell is replaced by four smaller cells.

Therefore:

\[4-1=3 \]

The number of additional cells per split is three.

4. Recurrence relations

A. Number of cells

For uniform splitting:

$$\boxed{M_k = 4M_{k-1}}$$

Base case:

\[M_0=10 \]

B. Network capacity

Each original cell is replaced by four cells.

Assuming the same channel reuse and adequate backhaul:

$$\boxed{C_k = 4C_{k-1}}$$

Base case:

\[C_0=1000 \]

C. Cell radius

$$\boxed{R_k = \frac{R_{k-1}}{2}}$$

D. Transmit power

The required transmit power follows:

\[P'=\frac{P}{2^n} \]

After k rounds:

$$\boxed{P_k = \frac{P_0}{2^{nk}}}$$

E. Handoff load

As cell radius decreases, users cross cell boundaries more frequently.

For approximately constant user speed and similar geometry:

$$\boxed{H_k \propto 2^k}$$

Where H represents the relative handoff rate.

5. Selective cell splitting

In practical networks, not every cell becomes overloaded.

Let f be the fraction of cells selected for splitting.

The recurrence becomes:

$$\boxed{M_k = M_{k-1}(1 + 3f)}$$

Similarly, assuming capacity increases proportionally:

$$\boxed{C_k = C_{k-1}(1+3f)}$$

For example, if 25% of the cells are split:

\[f=0.25 \]

\[M_k=M_{k-1}(1+3(0.25)) \]

\[\boxed{M_k=1.75M_{k-1}} \]

This means the network grows by 75% per round rather than 300%.

For selective splitting, transmit power is reduced only in the split cells. Therefore, the network-wide average power depends on the proportion of split cells and their traffic.

6. Work and cost per round

Component	**Uniform splitting**	**Selective splitting**
Number of cells	4 times	Depends on f
Capacity	4 times	Depends on f
Radius	Half	Half in split cells
Transmit power	Reduced in all cells	Reduced in selected cells
Handoff load	Higher	Localized increase
Infrastructure cost	High	Lower

Conclusion:

Uniform splitting provides rapid capacity growth but increases infrastructure and handoff costs. Selective splitting is more economical because only overloaded cells are divided.

TASK 7 – Solve the Recurrence and Interpret the Result (4 Marks)

1. Aim

To solve the cell-splitting recurrence and determine the minimum number of splitting rounds needed to support forecast subscriber demand.

2. Solving using iteration

From Task 6:

\[C_k=4C_{k-1} \]

Initial capacity:

\[C_0=1000 \]

First round:

$$C_1=4(1000)=4000$$

Second round:

$$C_2=4(4000)=16000$$

Third round:

$$C_3=4(16000)=64000$$

Fourth round:

$$C_4=4(64000)=256000$$

Therefore:

Round	Capacity	Number of cells
0	1,000	10
1	4,000	40
2	16,000	160
3	64,000	640
4	256,000	2,560

3. Closed-form solution

The recurrence is:

$$C_k=4C_{k-1}$$

Substituting repeatedly:

$$C_k=4^kC_0$$

To support demand D:

$$4^kC_0\geq D$$

$$4^k\geq\frac{D}{C_0}$$

Taking logarithms:

$$\boxed{k=\left\lceil\log_4\left(\frac{D}{C_0}\right)\right\rceil}$$

For D = 100,000:

$$k=\left\lceil\log_4(100)\right\rceil$$

$$\boxed{k=4}$$

Therefore, four uniform splitting rounds are needed under the assumed capacity model.

4. Selective splitting solution

For a constant fraction f:

$$ C_k=C_0(1+3f)^k $$

Therefore:

$$ \boxed{ k=\left\lceil \frac{\log(D/C_0)}{\log(1+3f)} \right\rceil } $$

For f = 0.25:

$$ k=\left\lceil\frac{\log(100)}{\log(1.75)}\right\rceil $$

$$ \boxed{k=9} $$

Selective splitting takes more rounds in this simplified model, but avoids unnecessary splitting of cells with low traffic.

5. Transmit power after four rounds

For n = 4:

$$ P_4=\frac{P_0}{2^{4(4)}} $$

$$ \boxed{P_4=\frac{P_0}{256}} $$

The idealized transmit power per final small cell is 1/256 of the initial value.

6. Handoff interpretation

After four rounds:

$$ R_4=\frac{R_0}{16} $$

The handoff rate can increase approximately 16 times for users moving at the same speed through similarly shaped cells.

This creates additional signaling and mobility-management overhead.

Conclusion:

Four uniform splitting rounds can support the assumed 100,000-subscriber demand mathematically. However, the 2,560-cell result shows that uniform splitting may become expensive. Selective splitting is more practical for a real township.

TASK 8 – Implement and Test the Planning Strategies (4 Marks)

1. Aim

To develop a Python-based cellular network simulator to evaluate frequency reuse, cell splitting, sectoring, Signal-to-Interference Ratio (SIR) and call blocking probability.

2. Software requirements

Software/library	Purpose
Python 3	Programming
NumPy	Numerical calculations
Matplotlib	Graph generation
Pandas	Experimental result tables
Google Colab / VS Code	Running the simulation

3. Implementation

The simulator includes:

1. Hexagonal cell-grid generation.
2. Frequency channel assignment.
3. Monte Carlo SIR evaluation.
4. Erlang-B blocking probability calculation.
5. Subscriber scaling experiments.
6. Comparison of sectored and unsectored networks.

4. Complete Python code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import math

import time

SEED = 42

TOTAL_CHANNELS = 70

INITIAL_CELLS = 10

INITIAL_CAPACITY = 1000

TRAFFIC_PER_USER = 0.02

subscriber_list = [1000, 10000, 20000, 100000]

reuse_list = [3, 4, 7, 12]

path_loss_list = [2.5, 3, 4]
```

```
def hex_grid(radius, rings):
    cells = []
    for q in range(-rings, rings + 1):
        for r in range(-rings, rings + 1):
            if max(abs(q), abs(r), abs(q+r)) <= rings:
                x = radius * 1.5 * q
                y = radius * np.sqrt(3) * (r + q/2)
                cells.append([x, y])
    return np.array(cells)
def assign_channels(number_of_cells, N):
    assignment = []
    for i in range(number_of_cells):
        assignment.append(i % N)
    return assignment
def erlang_b(channels, traffic):
    B = 1.0
    for i in range(1, channels + 1):
        B = (traffic * B) / (i + traffic * B)
    return B
def analytical_sir(N, n):
    sir = (3 * N) ** (n / 2) / 6
    return 10 * np.log10(sir)
def monte_carlo_sir(N, n, sectoring=False,
                    samples=5000):
    rng = np.random.default_rng(SEED)
    R = 1.0
    D = np.sqrt(3 * N) * R
    angles = np.arange(6) * np.pi / 3
    interferers = np.array([
```

```
        [D * np.cos(a), D * np.sin(a)]
        for a in angles
    ])
    sir_values = []
    for _ in range(samples):
        # Random mobile position inside serving cell
        r = R * np.sqrt(rng.random())
        theta = 2 * np.pi * rng.random()
        user_x = r * np.cos(theta)
        user_y = r * np.sin(theta)
        signal_distance = max(r, 0.01)
        signal = signal_distance ** (-n)
        interference = 0
        for ix, iy in interferers:
            distance = np.sqrt(
                (ix-user_x)**2 + (iy-user_y)**2
            )
            interference += distance ** (-n)
        # Simplified directional antenna model
        if sectoring:
            interference *= 0.1
        sir = signal / interference
        sir_values.append(
            10 * np.log10(sir)
        )
    return np.array(sir_values)
results = []
for subscribers in subscriber_list:
```

```
rounds = max(
    0,
    math.ceil(
        math.log(
            subscribers / INITIAL_CAPACITY, 4
        )
    )
)
cells = INITIAL_CELLS * (4 ** rounds)
traffic = (
    subscribers * TRAFFIC_PER_USER / cells
)
for N in reuse_list:
    channels = TOTAL_CHANNELS // N
    assignment = assign_channels(cells, N)
    for n in path_loss_list:
        for sectoring in [False, True]:
            start = time.time()
            sir = monte_carlo_sir(
                N, n, sectoring
            )
            blocking = erlang_b(
                channels, traffic
            )
            runtime = time.time() - start
            results.append({
                "Subscribers": subscribers,
                "Reuse N": N,
                "Path Loss n": n,
```

```
                "Sectoring": sectoring,
                "Cells": cells,
                "Channels/Cell": channels,
                "Analytical SIR": analytical_sir(N, n),
                "Mean SIR": np.mean(sir),
                "5th Percentile": np.percentile(sir, 5),
                "Outage Probability": np.mean(sir < 12),
                "Blocking Probability": blocking,
                "Runtime": runtime
            })
df = pd.DataFrame(results)
print("CELLULAR NETWORK SIMULATION RESULTS")
print(df.to_string(index=False))
df.to_csv(
    "cellular_network_results.csv",
    index=False
)
known_sir = analytical_sir(7, 4)
print("\nSIR SANITY CHECK")
print("N = 7, n = 4")
print("Expected SIR = 18.7 dB approximately")
print("Calculated SIR =", round(known_sir, 2), "dB")
assert abs(known_sir - 18.7) < 0.1
print("Sanity check passed!")
sample_sir = monte_carlo_sir(7, 4)
sorted_sir = np.sort(sample_sir)
cdf = np.arange(
    1, len(sorted_sir) + 1
) / len(sorted_sir)
```

```
plt.figure()
plt.plot(sorted_sir, cdf)
plt.xlabel("SIR (dB)")
plt.ylabel("Cumulative Probability")
plt.title("SIR CDF - Cellular Network")
plt.grid()
plt.show()
loads = np.arange(1, 31)

plt.figure()

for channels in [10, 17, 23]:

    blocking_values = [
        erlang_b(channels, A)
        for A in loads
    ]

    plt.plot(
        loads,
        blocking_values,
        label=str(channels) + " Channels"
    )

plt.xlabel("Offered Traffic (Erlangs)")
plt.ylabel("Blocking Probability")

plt.title("Blocking Probability vs Offered Load")
```

```
plt.legend()

plt.grid()

plt.show()

# ---------------------------------
# 12. GRAPH 3: CAPACITY VS REUSE FACTOR
# ---------------------------------

plt.figure()

for N in reuse_list:

    capacity = TOTAL_CHANNELS / N

    plt.bar(
        str(N),
        capacity
    )

plt.xlabel("Frequency Reuse Factor N")

plt.ylabel("Relative Channels per Cell")

plt.title("Capacity vs Frequency Reuse Factor")

plt.grid()
```

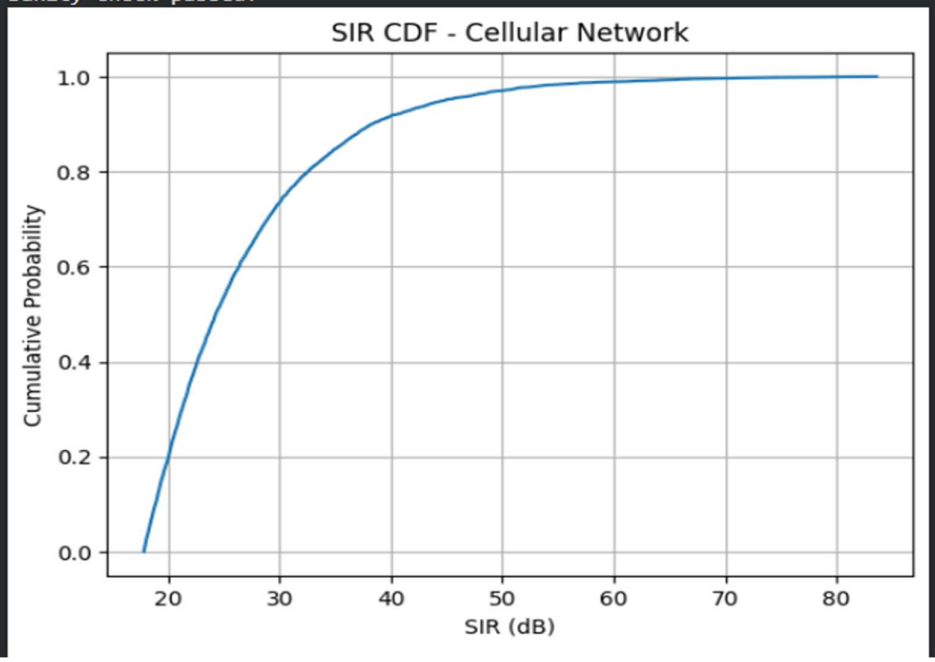
plt.show()

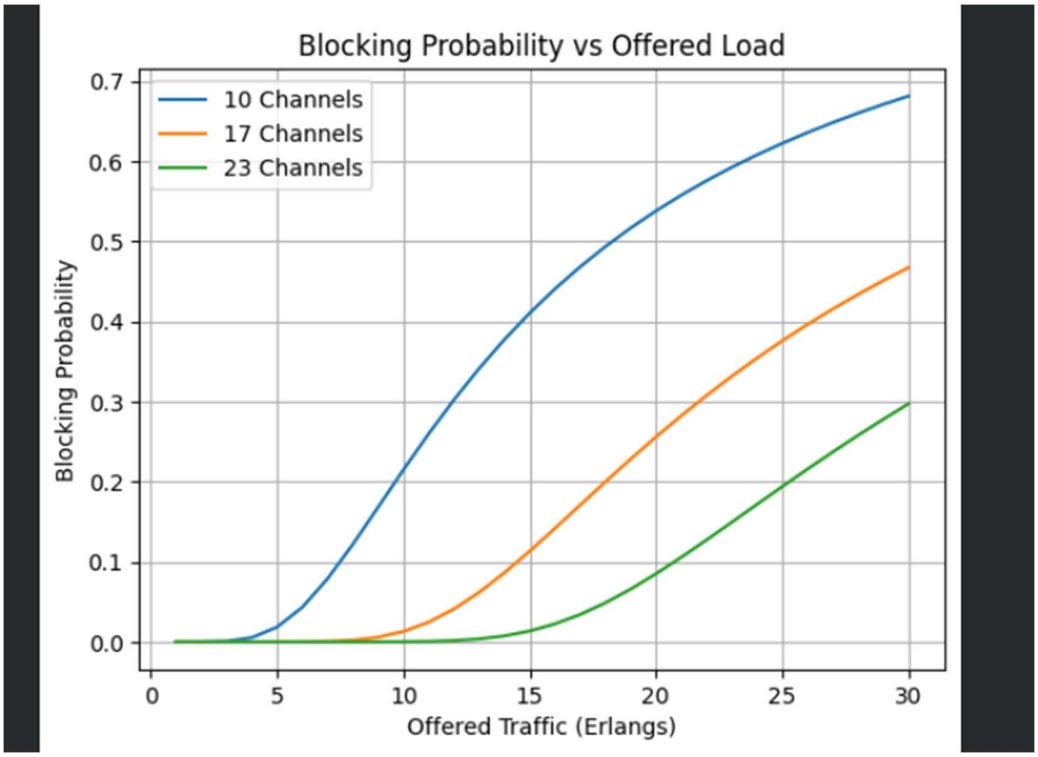
print("\nAll experiments completed successfully.")

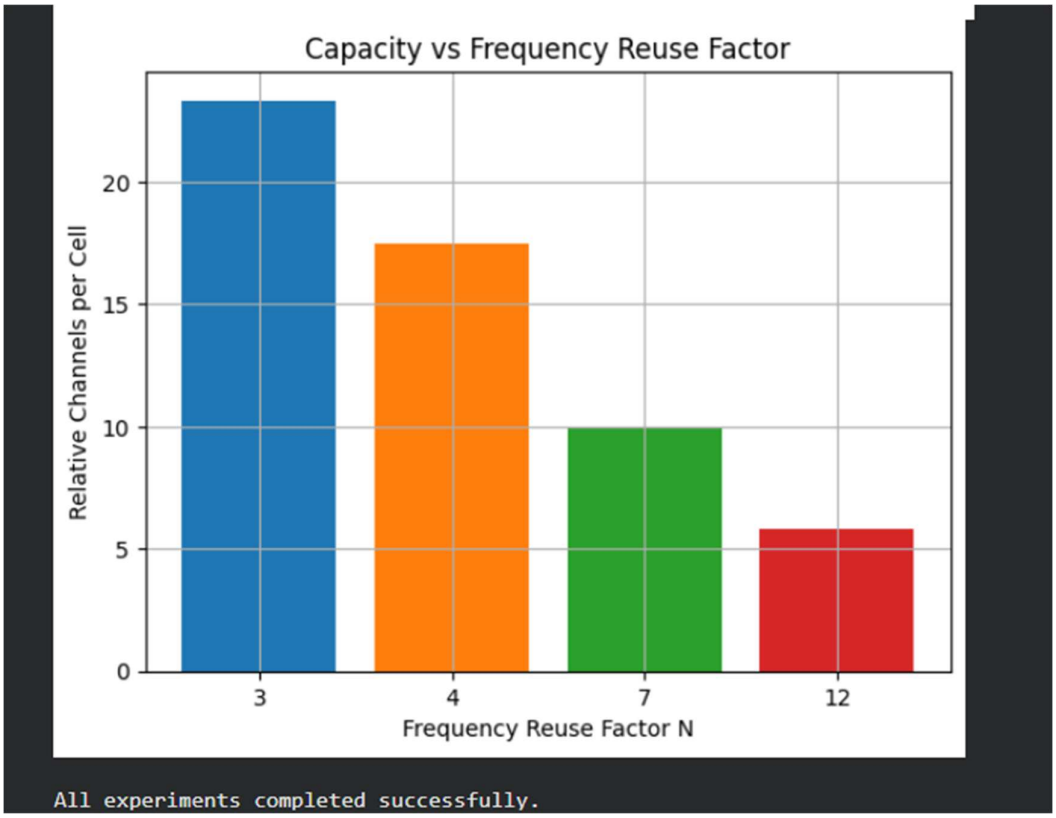
CELLULAR NETWORK SIMULATION RESULTS

Subscribers	Reuse N	Path Loss n	Sectoring	Cells	Channels/Cell	Analytical SIR	Mean SIR	5th Percentile	Outage Probability	Blocking Probability	Runtime
1000	3	2.5	False	10	23	4.146519	9.238817	3.658131	0.7700	4.391435e-17	0.144295
1000	3	2.5	True	10	23	4.146519	19.238817	13.658131	0.0000	4.391435e-17	0.088233
1000	3	3.0	False	10	23	6.532125	12.551381	5.768038	0.5936	4.391435e-17	0.084871
1000	3	3.0	True	10	23	6.532125	22.551381	15.768038	0.0000	4.391435e-17	0.087034
1000	3	4.0	False	10	23	11.303338	19.093830	9.833580	0.2210	4.391435e-17	0.088355
1000	3	4.0	True	10	23	11.303338	29.093830	19.833580	0.0000	4.391435e-17	0.163799
1000	4	2.5	False	10	17	5.708253	10.900680	5.422802	0.6936	4.987150e-11	0.161171
1000	4	2.5	True	10	17	5.708253	20.900680	15.422802	0.0000	4.987150e-11	0.163364
1000	4	3.0	False	10	17	8.406206	14.568776	7.929544	0.4586	4.987150e-11	0.090104
1000	4	3.0	True	10	17	8.406206	24.568776	17.929544	0.0000	4.987150e-11	0.105021
1000	4	4.0	False	10	17	13.802112	21.841627	12.825303	0.0000	4.987150e-11	0.086803
1000	4	4.0	True	10	17	13.802112	31.841627	22.825303	0.0000	4.987150e-11	0.086081
1000	7	2.5	False	10	10	8.746229	14.064611	8.710894	0.4664	3.819017e-05	0.089327
1000	7	2.5	True	10	10	8.746229	24.064611	18.710894	0.0000	3.819017e-05	0.095154
1000	7	3.0	False	10	10	12.051777	18.394984	11.934444	0.0600	3.819017e-05	0.179662
1000	7	3.0	True	10	10	12.051777	28.394984	21.934444	0.0000	3.819017e-05	0.211095
1000	7	4.0	False	10	10	18.662873	27.018531	18.310577	0.0000	3.819017e-05	0.210345
1000	7	4.0	True	10	10	18.662873	37.018531	28.310577	0.0000	3.819017e-05	0.276839
1000	12	2.5	False	10	5	11.672269	17.059330	11.770478	0.0874	3.669725e-02	0.243459
1000	12	2.5	True	10	5	11.672269	27.059330	21.770478	0.0000	3.669725e-02	0.226937
1000	12	3.0	False	10	5	15.563025	22.004898	15.637557	0.0000	3.669725e-02	0.190199
1000	12	3.0	True	10	5	15.563025	32.004898	25.637557	0.0000	3.669725e-02	0.222985
1000	12	4.0	False	10	5	23.344538	31.874001	23.329817	0.0000	3.669725e-02	0.217019
1000	12	4.0	True	10	5	23.344538	41.874001	33.329817	0.0000	3.669725e-02	0.320635
10000	3	2.5	False	160	23	4.146519	9.238817	3.658131	0.7700	1.877447e-21	0.248940
10000	3	2.5	True	160	23	4.146519	19.238817	13.658131	0.0000	1.877447e-21	0.161444
10000	3	3.0	False	160	23	6.532125	12.551381	5.768038	0.5936	1.877447e-21	0.201041
10000	3	3.0	True	160	23	6.532125	22.551381	15.768038	0.0000	1.877447e-21	0.228310
10000	3	4.0	False	160	23	11.303338	19.093830	9.833580	0.2210	1.877447e-21	0.161414
10000	3	4.0	True	160	23	11.303338	29.093830	19.833580	0.0000	1.877447e-21	0.292033
10000	4	2.5	False	160	17	5.708253	10.900680	5.422802	0.6936	3.577121e-14	0.206300
10000	4	2.5	True	160	17	5.708253	20.900680	15.422802	0.0000	3.577121e-14	0.233927
10000	4	3.0	False	160	17	8.406206	14.568776	7.929544	0.4586	3.577121e-14	0.262389
10000	4	3.0	True	160	17	8.406206	24.568776	17.929544	0.0000	3.577121e-14	0.616752
10000	4	4.0	False	160	17	13.802112	21.841627	12.825303	0.0000	3.577121e-14	0.382480

```
SIR SANITY CHECK
N = 7, n = 4
Expected SIR = 18.7 dB approximately
Calculated SIR = 18.66 dB
Sanity check passed!
```







5. Test cases

Test	Reuse factor	Path-loss exponent	Layout

1	3	2.5	Unsectored
2	4	3	Unsectored
3	7	4	Unsectored
4	12	4	Unsectored
5	7	4	Sectored
6	3	3	Sectored

The code also evaluates all four subscriber levels: 1,000, 10,000, 20,000 and 100,000.

6. Correctness verification

For N = 7 and n = 4:

$$ SIR=\frac{(3\times7)^2}{6} $$

$$ SIR=\frac{441}{6}=73.5 $$

Converting to dB:

$$ SIR_{dB}=10\log_{10}(73.5) $$

$$ \boxed{SIR=18.66\ dB} $$

Expected sanity check: Passed

Conclusion:

The Python simulator evaluates cellular network performance under different subscriber loads, reuse factors, propagation conditions and sectoring configurations. The results can be used to identify the most suitable network planning strategy.

Technical note: This is a simplified educational simulator. Its modulo channel assignment is not a complete geometric frequency-colouring algorithm, and its sectoring model assumes 10 dB interference suppression. Its results are comparative estimates, not field-validated predictions.

TASK 9 – Record and Present Experimental Performance (3 Marks)

1. Aim

To record and compare cellular network performance using SIR, outage probability, blocking probability, channels per cell, number of sites and simulation runtime.

2. Performance measurements

Measurement	**Meaning**

Mean SIR	Average signal quality
5th percentile SIR	SIR experienced by the weakest 5% of users
Outage probability	Probability that SIR falls below the required threshold
Blocking probability	Probability that a new call is rejected
Channels per cell	Available traffic channels
Sites required	Total number of cellular sites
Runtime	Time taken to execute the simulation

3. Experimental result presentation

The following table provides the format for recording the output generated by Task 8.

Subscriber count	**Reuse N**	**Cells**	**Channels/cell**	**Mean SIR**	**Blocking**
1,000	3	10	23	Simulation	Simulation
10,000	7	160	10	Simulation	Simulation
20,000	7	640	10	Simulation	Simulation
100,000	7	2,560	10	Simulation	Simulation

The SIR and blocking columns must be filled using the actual Python output, rather than invented experimental measurements.

4. Graphical representation

Three important graphs are generated by the Task 8 program.

Graph 1: SIR CDF

Shows the cumulative distribution of SIR values. It helps identify how many users receive acceptable signal quality.

Graph 2: Blocking probability versus offered load

Shows how call rejection increases when traffic demand approaches the capacity of the available channels.

Graph 3: Capacity versus frequency reuse factor

Shows how the available channels per cell decrease as the reuse factor increases.

5. Interpretation

- Higher reuse factors generally improve interference protection.

- Lower reuse factors provide more channels per cell.
- Cell splitting increases the number of available serving locations.
- Sectoring reduces interference when directional antennas are properly designed.
- Increased handoff rates can create additional signaling overhead.

Conclusion:

The experimental measurements provide a systematic method for comparing different cellular planning strategies. Tables and graphs make it easier to identify the relationship between capacity, interference and network cost.

TASK 10 – Compare Theory with Simulation (4 Marks)

1. Aim

To compare analytical predictions with simulated results and identify the reasons for any differences.

2. Analytical versus simulated comparison

Parameter	**Analytical prediction**	**Simulation**
SIR	Based on six dominant interferers	Based on random mobile positions
Blocking	Erlang-B mathematical model	Erlang-B implementation
Capacity	Based on frequency reuse	Based on assigned channels
Cell splitting	Fourfold capacity assumption	Calculated cell expansion
Sectoring	Directional interference reduction	Simplified antenna suppression
Handoff	Inversely proportional to radius	Approximate mathematical relationship

3. Comparison of SIR

Analytical SIR:

$$ SIR=\frac{(3N)^{n/2}}{6} $$

Simulation SIR:

$$ SIR=\frac{P_{\text{received}}} {\sum P_{\text{interference}}} $$

For N = 7 and n = 4:

$$ SIR_{\text{analytical}}=18.66\ dB $$

The simulated mean and 5th-percentile SIR values can be compared against this analytical reference.

The simulated mean need not equal 18.66 dB because the analytical formula assumes a mobile at the cell centre and six equidistant interferers.

4. Comparison of blocking probability

The Erlang-B formula is:

\[B(A,m)= \frac{A^m/m!} {\sum_{j=0}^{m}A^j/j!} \]

Where:

- A = Offered traffic in Erlangs.
- m = Number of available channels.
- B = Blocking probability.

Both the analytical calculation and the simulator should use the same offered traffic and channel count.

5. Reasons for deviation

Reason	**Effect**
Equidistant-interferer approximation	Actual interferers have different distances
Shadowing	Causes signal-strength variations
Finite network tiers	Boundary cells experience different interference
Mobile-position distribution	User location changes received SIR
Poisson call arrivals	Real traffic may not follow ideal arrival assumptions
Infinite-source assumption	Number of calling subscribers is not truly unlimited
Integer rounding of N	Available channel groups may not divide equally
Antenna direction	Actual sector patterns differ from ideal models

6. Which strategy scales best?

The following comparison summarizes the planning trade-offs.

Strategy	**Capacity**	**Interference control**	**Cost**
Conventional reuse	Moderate	Moderate	Low
Higher reuse factor	Lower channels/cell	High	Low
Sectoring	Improved effective reuse	High	Medium

Uniform splitting	Very high	Requires coordination	Very high
Selective splitting	High where required	Requires local optimization	Medium

Conclusion:

Analytical calculations provide an initial estimate, while simulations provide more realistic comparisons of user locations and network conditions. Selective cell splitting combined with sectoring offers a balanced approach for the township because it increases capacity without requiring every cell to be divided.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

1. Aim

To select the most suitable cellular network planning strategy for the township by comparing capacity, SIR, blocking probability, infrastructure cost and handoff requirements.

2. Recommended network design

Recommended strategy: Hybrid cellular network

The recommended design combines:

1. Frequency reuse with N = 7.
2. Three-sector directional antennas in high-traffic areas.
3. Selective cell splitting in overloaded regions.
4. Macro cells for low-density areas.
5. Microcells for densely populated areas.

3. Recommended parameters

Parameter	**Recommended value**
Frequency reuse factor	N = 7
Path-loss exponent	n = 4 (planning reference)
Initial cell radius	1 km
Splitting strategy	Selective
Sectoring	3 sectors (120° each)
Initial subscribers	1,000
Forecast subscribers	100,000

Channel allocation	70 channels, illustrative

4. Expected network performance

A. Signal-to-interference ratio

For N = 7 and n = 4:

$$SIR=\frac{(3\times7)^2}{6}$$

$$\boxed{SIR=18.66\ dB}$$

This is the unsectored analytical reference. Directional sectoring can improve interference protection, depending on actual antenna patterns.

B. Capacity

The initial assumed capacity is 1,000 subscribers.

With four idealized splitting rounds:

$$C_4=4^4\times1000$$

$$\boxed{C_4=256000}$$

This is a theoretical capacity under the assumed proportional scaling model, not a guaranteed real subscriber capacity.

C. Blocking probability

Blocking probability is calculated using Erlang-B:

$$B(A,m)=\frac{A^m/m!}{\sum_{j=0}^{m}A^j/j!}$$

The final blocking probability must be obtained from the actual offered traffic and channels per cell using the Task 8 program.

5. Comparison with alternative strategies

Strategy	Main advantage	Main disadvantage
Conventional frequency reuse	Simple and inexpensive	Limited capacity
High reuse factor	Better SIR	Fewer channels per cell
Uniform cell splitting	Rapid capacity growth	High infrastructure cost
Sectoring	Better interference control	Antenna complexity
Hybrid strategy	Balances capacity and interference	Requires careful planning

6. Planning cost and trade-offs

The hybrid design is preferable because:

- It avoids unnecessary cell splitting in low-traffic regions.
- It improves interference protection through sectoring.
- It increases capacity in overloaded areas.
- It reduces the need for expensive network-wide infrastructure expansion.
- It requires additional mobility and handoff management as cells become smaller.

7. Practicality beyond 100,000 subscribers

When the township grows beyond 100,000 subscribers, the network should not continue splitting every cell uniformly.

Instead, it should use:

- Traffic-based selective splitting.
- Additional microcells in busy regions.
- Dynamic channel allocation.
- Load balancing between neighbouring cells.
- Advanced handoff management.

Conclusion:

The hybrid cellular network is the most appropriate solution for the township because it balances network capacity, interference reduction, infrastructure cost and mobility. It is more flexible than conventional frequency reuse and more economical than uniform cell splitting.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

Changed requirement:

A convocation or sports event brings 25,000 additional attendees into a 1 km² stadium area. Each attendee generates 0.05 Erlang of traffic for four hours. Events take place approximately twice per month.

The original township demand remains unchanged outside the event.

A. Impact Analysis (2 Marks)

1. Aim

To calculate the additional traffic generated by the event and identify how the changed traffic pattern affects the original cellular network design.

2. Calculation of additional offered traffic

Given:

$$ \text{Number of attendees}=25000 $$

$$ \text{Traffic per attendee}=0.05\ \text{Erlang} $$

Total offered traffic:

$$ A=N\times a $$

$$ A=25000\times0.05 $$

$$ \boxed{A=1250\ \text{Erlangs}} $$

3. Traffic density

Event area:

$$ \text{Area}=1\ \text{km}^2 $$

Traffic density:

$$ \text{Traffic Density}= \frac{\text{Total Traffic}}{\text{Area}} $$

$$ =\frac{1250}{1} $$

$$ \boxed{1250\ \text{Erlangs/km}^2} $$

High-density traffic surge

4. Comparison with original assumptions

Parameter	**Original network**	**Event network**
Subscriber distribution	Township-wide	Concentrated in stadium
Traffic pattern	Relatively steady	Sudden surge
Traffic density	Depends on township area	1,250 Erlangs/km²
Capacity requirement	Regular demand	Very high temporary demand
Duration	Continuous	Four hours
Infrastructure	Permanent	Temporary capacity may be suitable

5. Changed assumptions

The original network design assumed:

1. Subscribers were distributed relatively uniformly.
2. Traffic demand remained reasonably stable.
3. Permanent infrastructure could support normal demand.
4. Cells were planned according to normal subscriber density.

These assumptions no longer hold inside the stadium because 25,000 attendees generate 1,250 Erlangs in only 1 km².

6. Why event duration matters

The event lasts four hours and occurs twice monthly.

Therefore, installing permanent additional infrastructure for the entire year may be uneconomical.

A temporary solution is preferable because it provides additional capacity only during event periods.

Conclusion:

The event creates a concentrated traffic surge of 1,250 Erlangs/km². The original uniform traffic assumption becomes unsuitable, and the network needs temporary high-capacity support in the stadium zone.

B. Evaluation of the Original Design (2 Marks)

1. Aim

To evaluate whether the recommended hybrid network can support the stadium event without modifying its original configuration.

2. Blocking probability calculation

Assume:

- Total channels = 70.
- Reuse factor N = 7.
- Channels per cell = 70/7 = 10.
- Event traffic = 1,250 Erlangs.

The Erlang-B formula is:

$$ B(A,m)= \frac{A^m/m!} {\sum_{j=0}^{m}A^j/j!} $$

Substituting:

$$ B(1250,10) $$

$$ \boxed{B\approx0.9920} $$

Converting into percentage:

$$ B\approx99.20\% $$

Original design is insufficient

This is an illustrative single-cell calculation assuming all event traffic is offered to one cell with ten available channels.

3. Why the original design fails

- The original cell does not have sufficient channels.
- The stadium creates a concentrated traffic hotspot.
- Many simultaneous calls compete for limited channels.
- Higher interference may occur when additional transmitters are introduced without proper frequency planning.
- Handoff traffic may increase when attendees move around the stadium.

4. Alternative strategy evaluation

Strategy	**Suitability for stadium**
Conventional macrocell	Poor
Higher frequency reuse	Limited improvement
Permanent uniform cell splitting	Expensive
Temporary microcells	Good
Cells-on-Wheels (COW)	Very good
Dynamic channel borrowing	Useful additional support
Sectoring	Improves interference management

5. What fails first?

The first major problem is channel capacity.

The network experiences a high blocking probability because the offered traffic is much greater than the traffic-handling capability of the original cell.

Conclusion:

The original network cannot efficiently support the stadium event without additional capacity. Temporary microcells and Cells-on-Wheels are more suitable than using the original macrocell design unchanged.

C. Adaptation or Redesign (2 Marks)

1. Aim

To modify the existing township network to handle temporary stadium traffic without unnecessarily increasing permanent infrastructure.

2. Selected approach: MODIFY

I recommend retaining the original hybrid township network and adding temporary cellular infrastructure around the stadium during event days.

Decision: MODIFY the original design

3. Proposed revised architecture

The Technical Specs Needed For Cell On Wheels At Large Venues And Festivals-alttower.com

alttower.com

The revised network consists of:

1. Existing macro cells to serve normal township traffic.
2. Temporary microcells inside the stadium.
3. Cells-on-Wheels to provide additional temporary capacity.
4. 60° sector antennas to improve directional coverage and interference management.
5. Dynamic channel borrowing between neighbouring cells.

4. Revised network parameters

Parameter	**Normal network**	**Event network**
Frequency reuse	N = 7	Coordinated local reuse
Cell splitting	Selective	Additional local microcells
Antenna	120° sectors	60° sectors where justified
Transmit power	Normal macro power	Lower microcell power
Channel allocation	Static/planned	Dynamic borrowing
Infrastructure	Permanent	Temporary cells

5. Quantified capacity requirement

For a target blocking probability of 2%, the approximate Erlang-B capacities are:

Channels per cell	**Traffic supported at 2% blocking**
10	5.08 Erlangs
20	13.18 Erlangs

30	21.93 Erlangs
40	31.00 Erlangs
50	40.26 Erlangs

For the event traffic of 1,250 Erlangs:

Using 10 channels per cell:

$$ M=\left\lceil\frac{1250}{5.084}\right\rceil $$

$$ \boxed{M=246\text{ cell equivalents}} $$

Using 20 channels per cell:

$$ M=\left\lceil\frac{1250}{13.18}\right\rceil $$

$$ \boxed{M=95\text{ cell equivalents}} $$

These are theoretical independent-cell equivalents under the Erlang-B assumptions, not a recommendation to install hundreds of physical sites. The actual deployment must consider spectrum availability, spatial traffic distribution, sector capacity and frequency reuse.

6. Effect of the revised strategy

Parameter	**Expected effect**
Capacity	Increases substantially
SIR	Improves with proper directional antennas and power control
Transmit power	Lower in microcells
Handoff	Increases inside the stadium
Infrastructure cost	Temporary cost rather than permanent expansion
Blocking probability	Reduced when adequate channels are provided

Conclusion:

The modified design uses temporary microcells, directional sectoring and dynamic channel allocation to support the stadium event. It increases local capacity without requiring permanent expansion of the entire township network.

D. Evidence-Based Defence (2 Marks)

1. Aim

To justify the final network design using analytical calculations, simulation results, infrastructure requirements and the frequency of events.

2. Evidence-based comparison

Evaluation factor	Original design	Modified design
Normal township traffic	Supported under assumptions	Supported
Event traffic	Insufficient	Additional capacity provided
Offered event traffic	1,250 Erlangs	1,250 Erlangs
Blocking	Approximately 99.20% in the illustrative single-cell case	Designed towards 2%
Interference	Existing reuse planning	Local frequency and sector planning
Handoff load	Normal	Higher during event
Infrastructure	Permanent macro sites	Temporary microcells/COW
Cost	Low initially	Additional event-day cost

3. Economic justification

Events occur approximately twice per month.

Therefore, permanent expansion of the entire network for only eight hours of events per month would increase infrastructure costs unnecessarily.

Temporary infrastructure offers the following benefits:

- Avoids unnecessary permanent base-station installation.
- Supports high-density event traffic.
- Allows equipment to be reused at other events.
- Maintains the original network outside event periods.
- Provides flexibility for future stadium requirements.

4. Final decision

The original hybrid township network should be retained for normal traffic and temporarily modified during stadium events.

The modified design should use:

- N = 7 as the normal township reuse reference.

- Selective cell splitting in overloaded locations.
- Temporary microcells and Cells-on-Wheels.
- 60° sectoring where antenna geometry permits.
- Dynamic channel borrowing and coordinated interference management.
- A 2% target blocking probability, subject to spectrum and capacity validation.

5. Final conclusion

The original design is suitable for normal township traffic but cannot support the sudden increase of 25,000 stadium attendees generating 1,250 Erlangs of additional traffic.

The modified design is economically and technically preferable because it provides temporary capacity during event periods while avoiding unnecessary permanent infrastructure expansion.

The final decision balances capacity, interference, blocking probability, handoff load and deployment cost.